

# Corrosion RC Beam Optimizer

## Scientific Report - Full Pipeline

*Generated: April 15, 2026 - 22:46*

This report presents the complete results of the Corrosion RC Beam Optimizer pipeline. The study applies ML ensemble models and symbolic regression (PySR) to predict the residual flexural capacity of corroded RC beams, benchmarked against ACI 318-19.

L1 target:  $R2 \geq 0.9$  | L2 target:  $R2 \geq 0.972$

Data: 804 specimens | Split: 70/30

## 1. ACI 318-19 Benchmark

R2: 0.8839

RMSE: 8.2289

MAE: 5.0617

MAPE: 29.9

ratio\_mean: 1.1845

ratio\_std: 0.5874

ratio\_min: 0.2735

ratio\_max: 5.2832

underestimate\_pct: 48.5

n\_specimens: 804

## 2. Ensemble Model Results (Part 1)

Best Model: CatBoost

Test R2 = 0.9752

Test RMSE = 3.9819 kN.m

Test MAE = 2.1459 kN.m

Test CV% = 17.1%

Test SD/M = 0.1693

### 10-Fold CV (ALL 804 samples):

R2 = 0.9748

RMSE = 3.8362 kN.m

MAE = 1.9407 kN.m

CV% = 16.9%

SD/M = 0.1689

### 3. PySR Symbolic Regression (Part 2)

Winner: DIRECT

Equation:  $d*d*(b - (CSI*0.13916598 + \exp(-d_b + \rho_t)))*((f_c - 9.325594)*7.8713674e-7*RI + 1.259213$

Mmax R2 = 0.9480

Mmax RMSE = 5.5058 kN.m

Mmax MAE = 3.6083 kN.m

Mmax MAPE = 25.89%

CV% = 24.26%

SD/M = 0.2425

L1 broken = True

L2 broken = False

Ratio approach R2 = 0.9229

Direct approach R2 = 0.9480

### 4. Pareto Front Equations

C= 1 | Mmax R2=0.0708 | MAPE=89.0%  
C= 3 | Mmax R2=0.5040 | MAPE=185.4%  
C= 4 | Mmax R2=0.5162 | MAPE=165.7%  
C= 5 | Mmax R2=0.7893 | MAPE=50.3%  
C= 6 | Mmax R2=0.8614 | MAPE=39.6%  
C= 7 | Mmax R2=0.8620 | MAPE=42.8%  
C= 8 | Mmax R2=0.8803 | MAPE=40.9%  
C= 9 | Mmax R2=0.8897 | MAPE=35.3%  
C=11 | Mmax R2=0.9048 | MAPE=36.8%  
C=12 | Mmax R2=0.9164 | MAPE=28.8%  
C=13 | Mmax R2=0.9289 | MAPE=27.2%  
C=15 | Mmax R2=0.9377 | MAPE=26.8%  
C=17 | Mmax R2=0.9387 | MAPE=26.8%  
C=18 | Mmax R2=0.9391 | MAPE=26.5%  
C=19 | Mmax R2=0.9413 | MAPE=25.3%  
C=20 | Mmax R2=0.9422 | MAPE=25.5%  
C=21 | Mmax R2=0.9444 | MAPE=27.0%  
C=22 | Mmax R2=0.9462 | MAPE=25.4%  
C=24 | Mmax R2=0.9480 | MAPE=25.9% <<<

5. Figures Gallery

Fig10 Pareto Mmax Performance

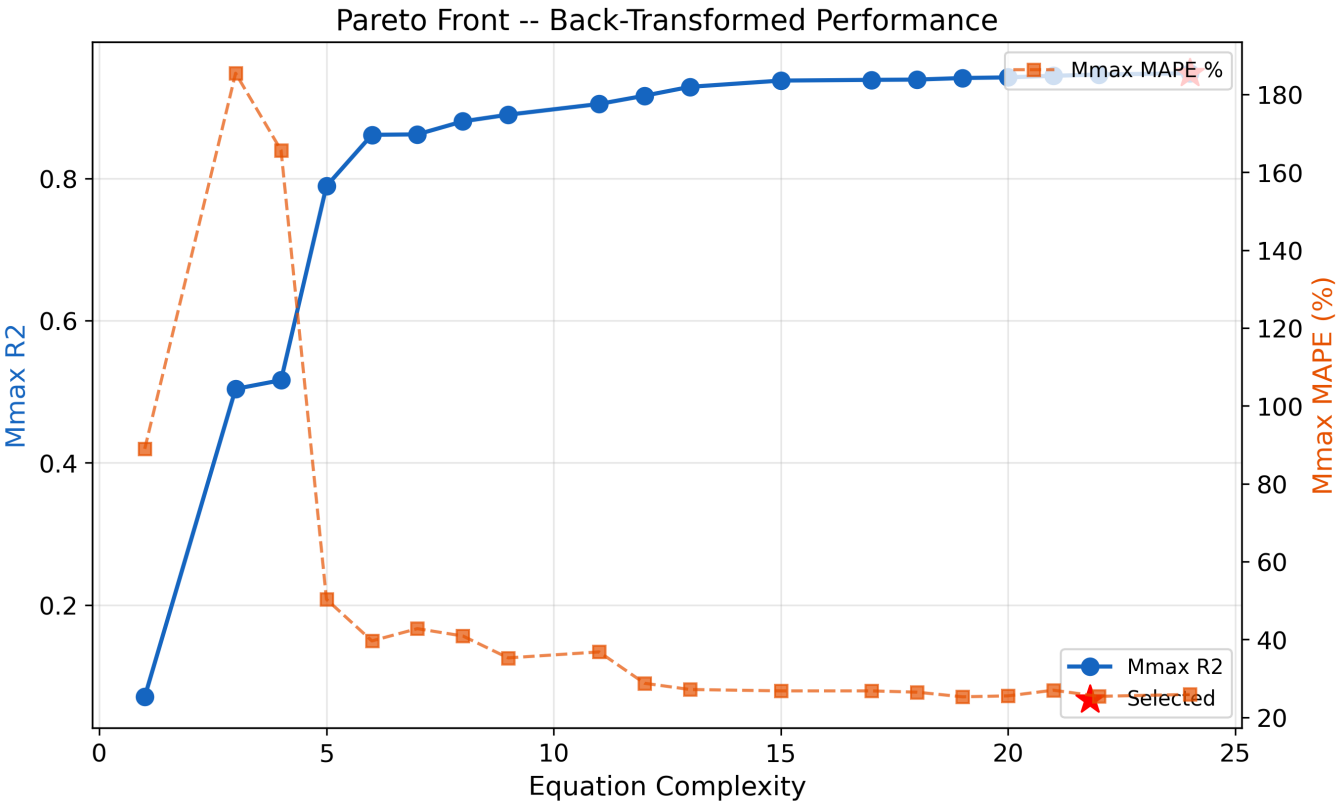


Fig11 Ratio Vs Eta

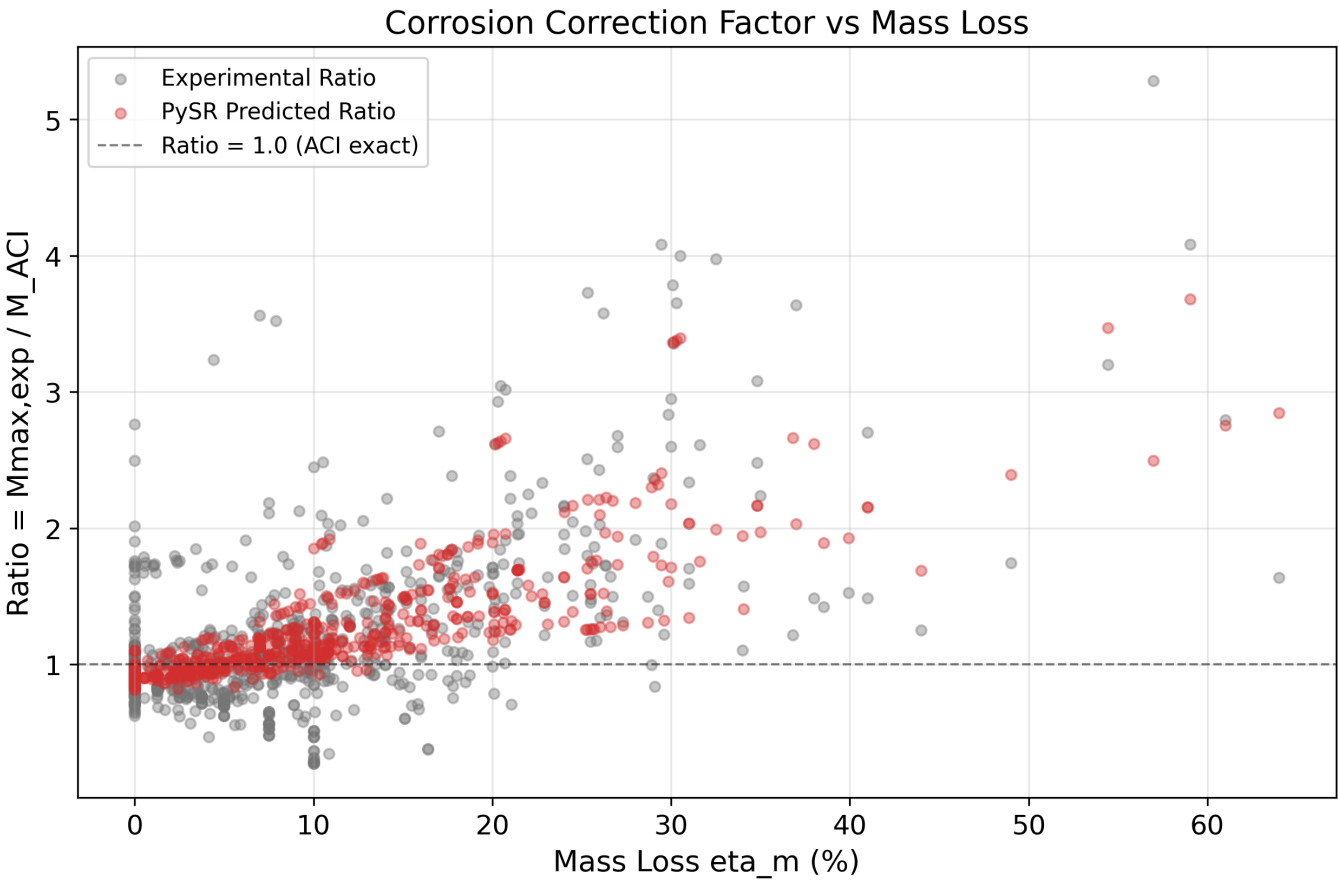


Fig1 Predicted Vs Experimental

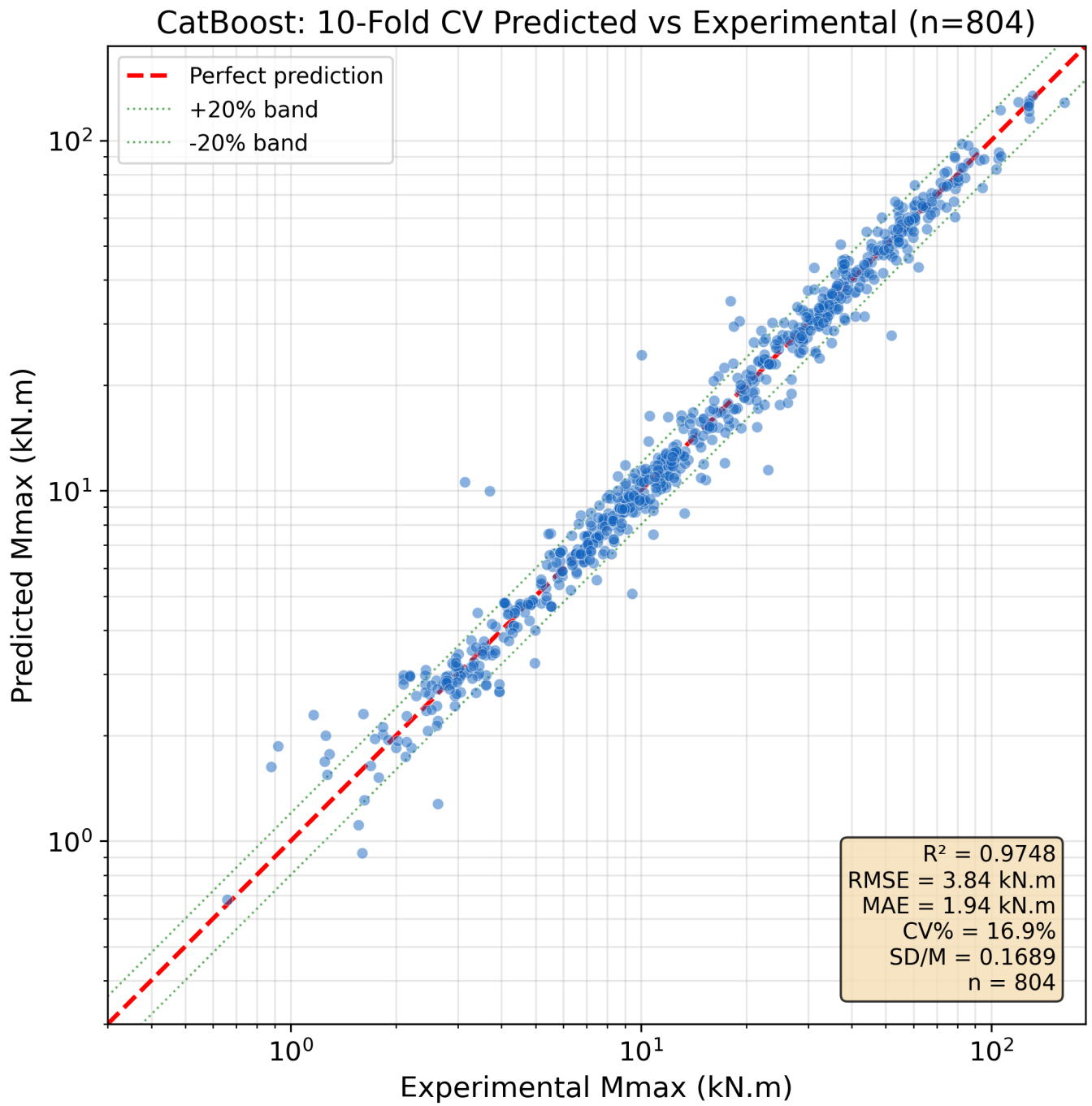


Fig1B Linear Scatter

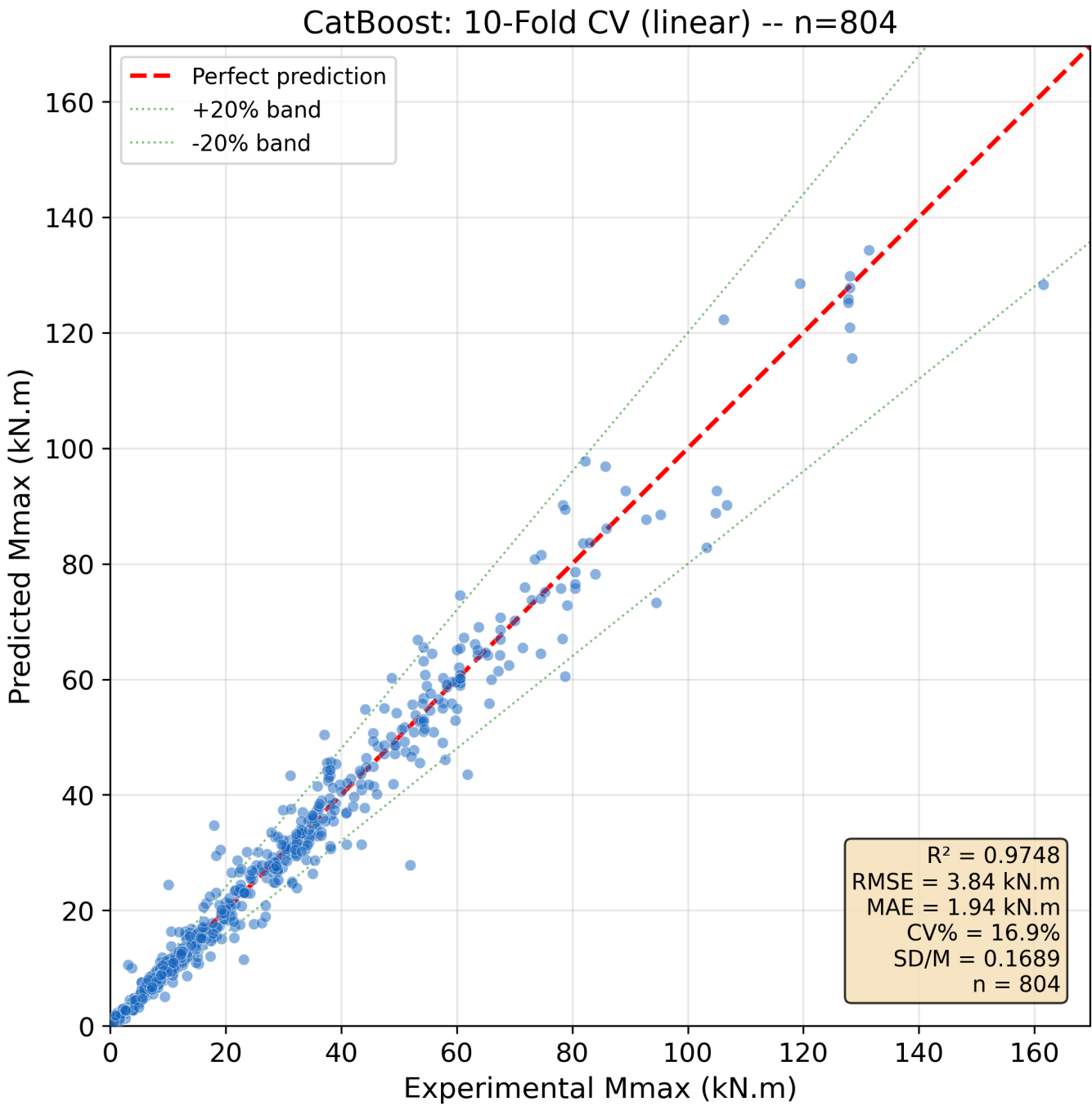


Fig2 Test Set Scatter

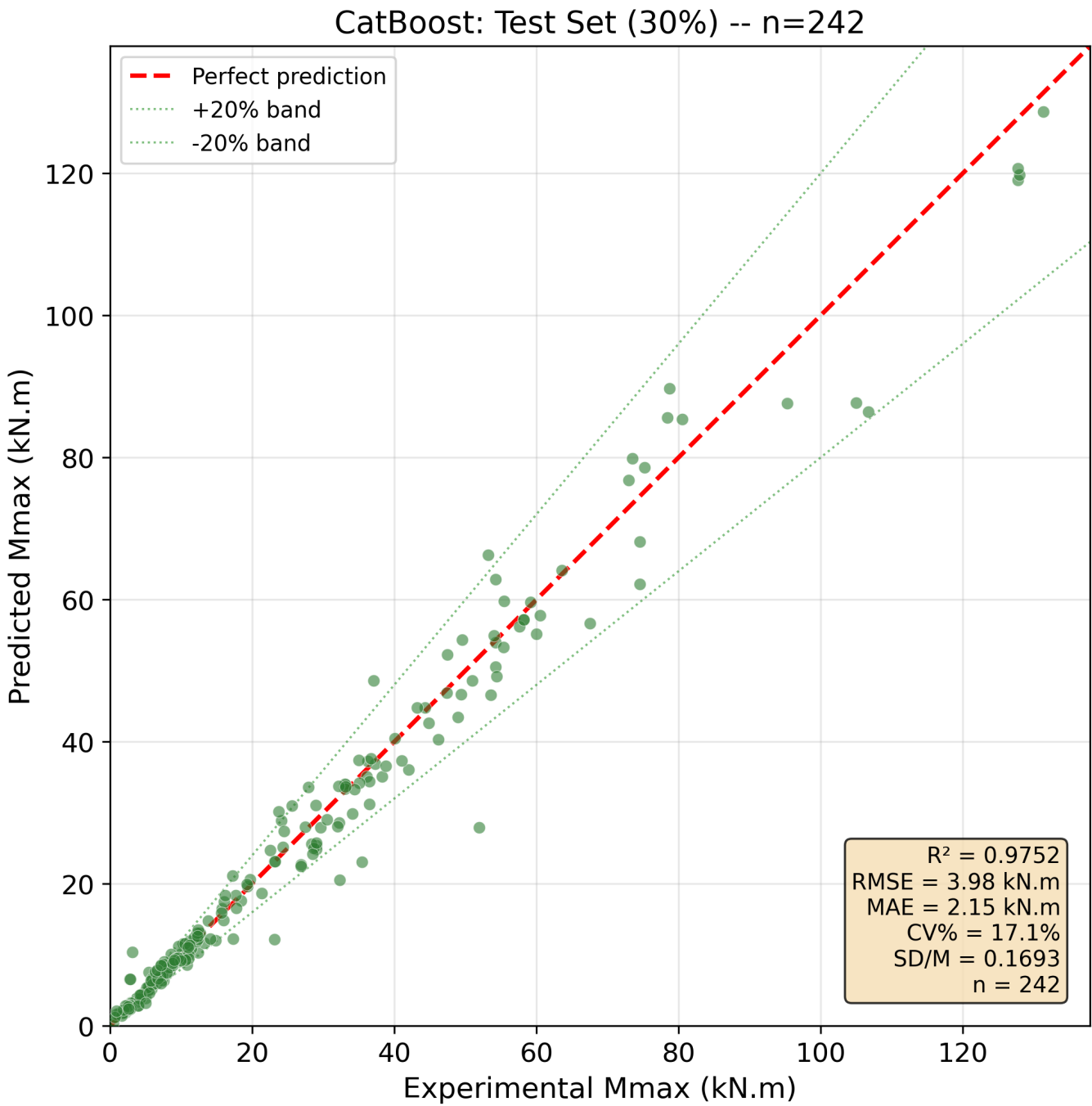


Fig3 Ensemble Vs Aci Scatter

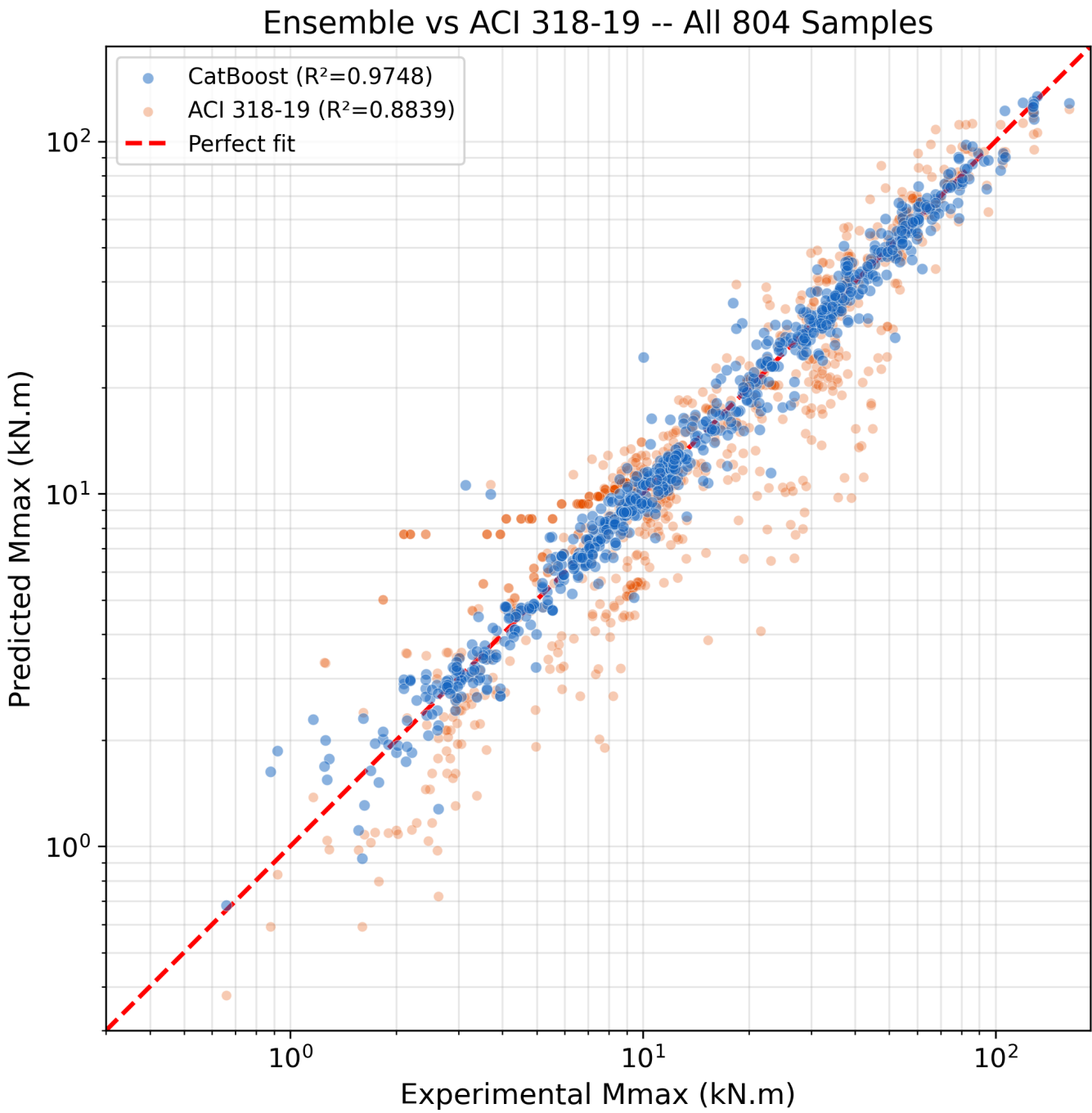




Fig4 Kfold Boxplot

10-Fold CV:  $R^2 = 0.9741 \pm 0.0094$

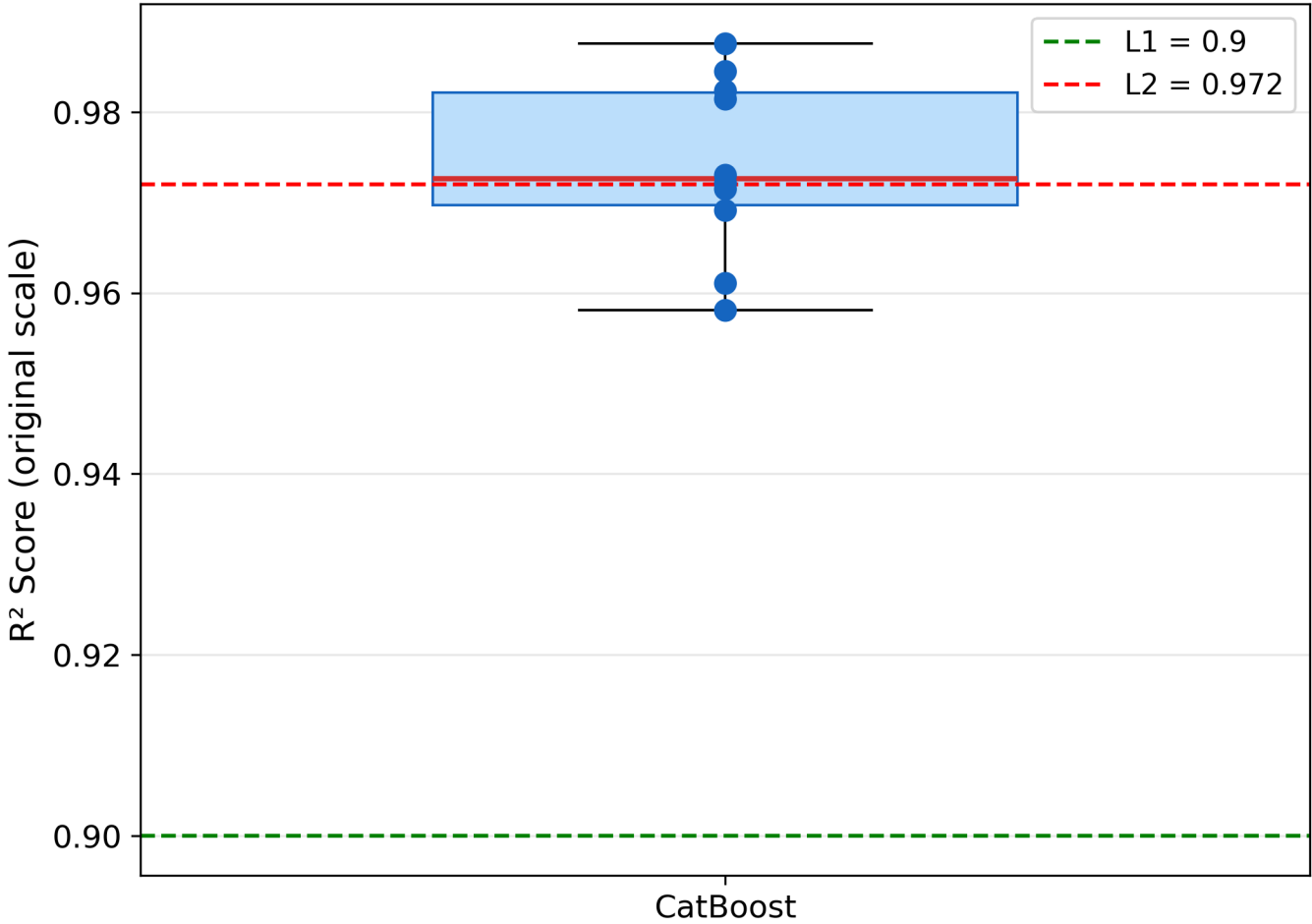


Fig5 Error Distribution

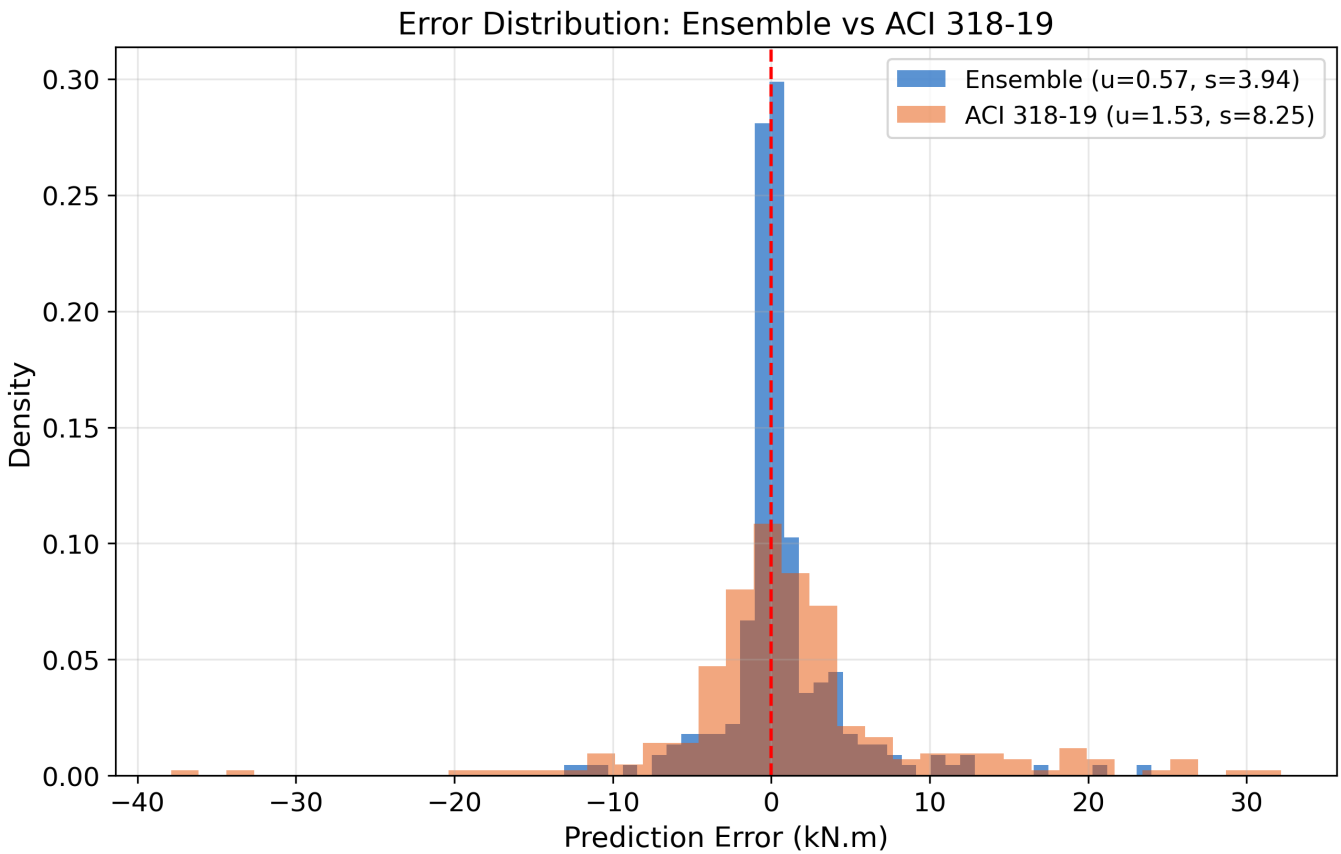


Fig6 Model Comparison

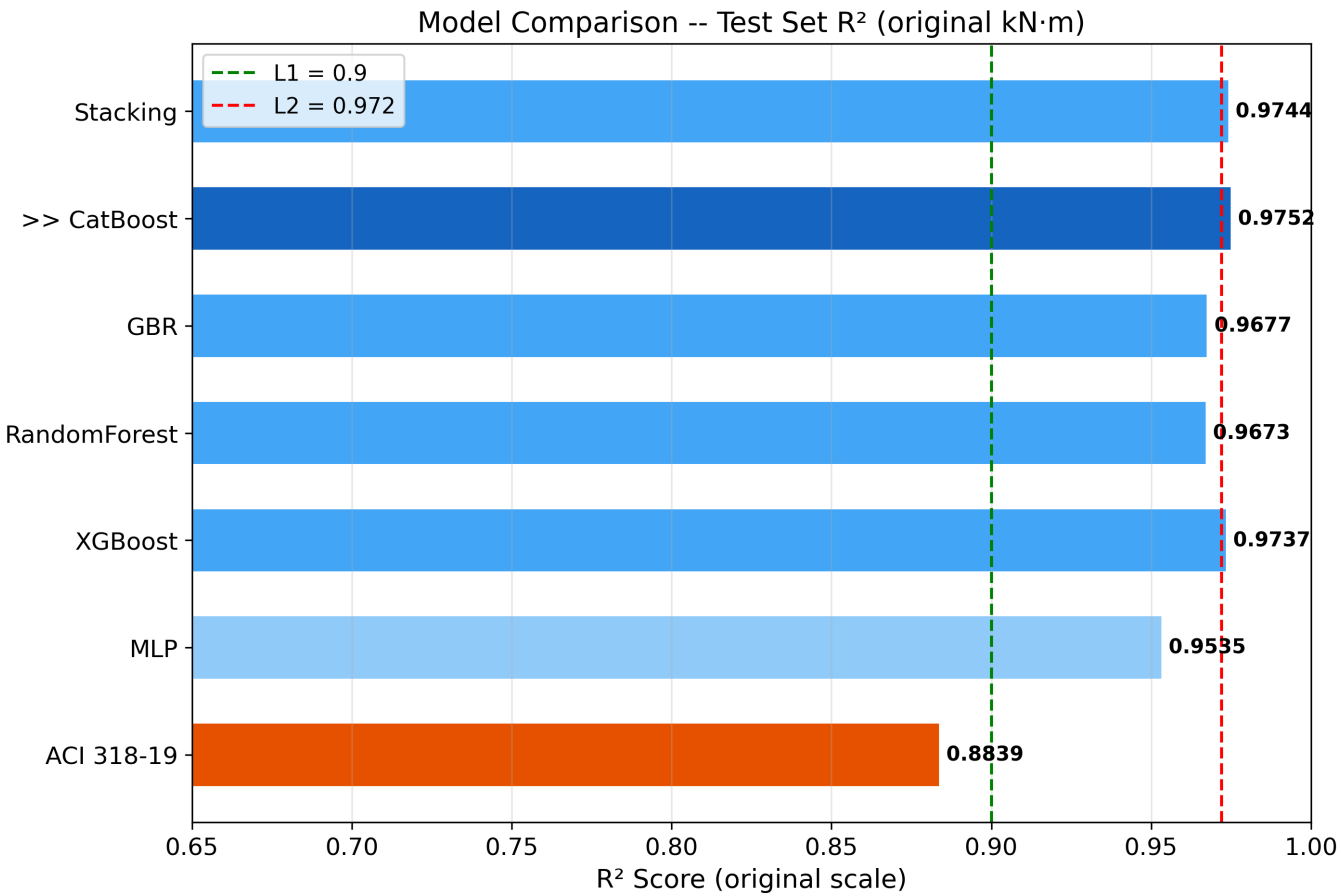


Fig7 Taylor Diagram

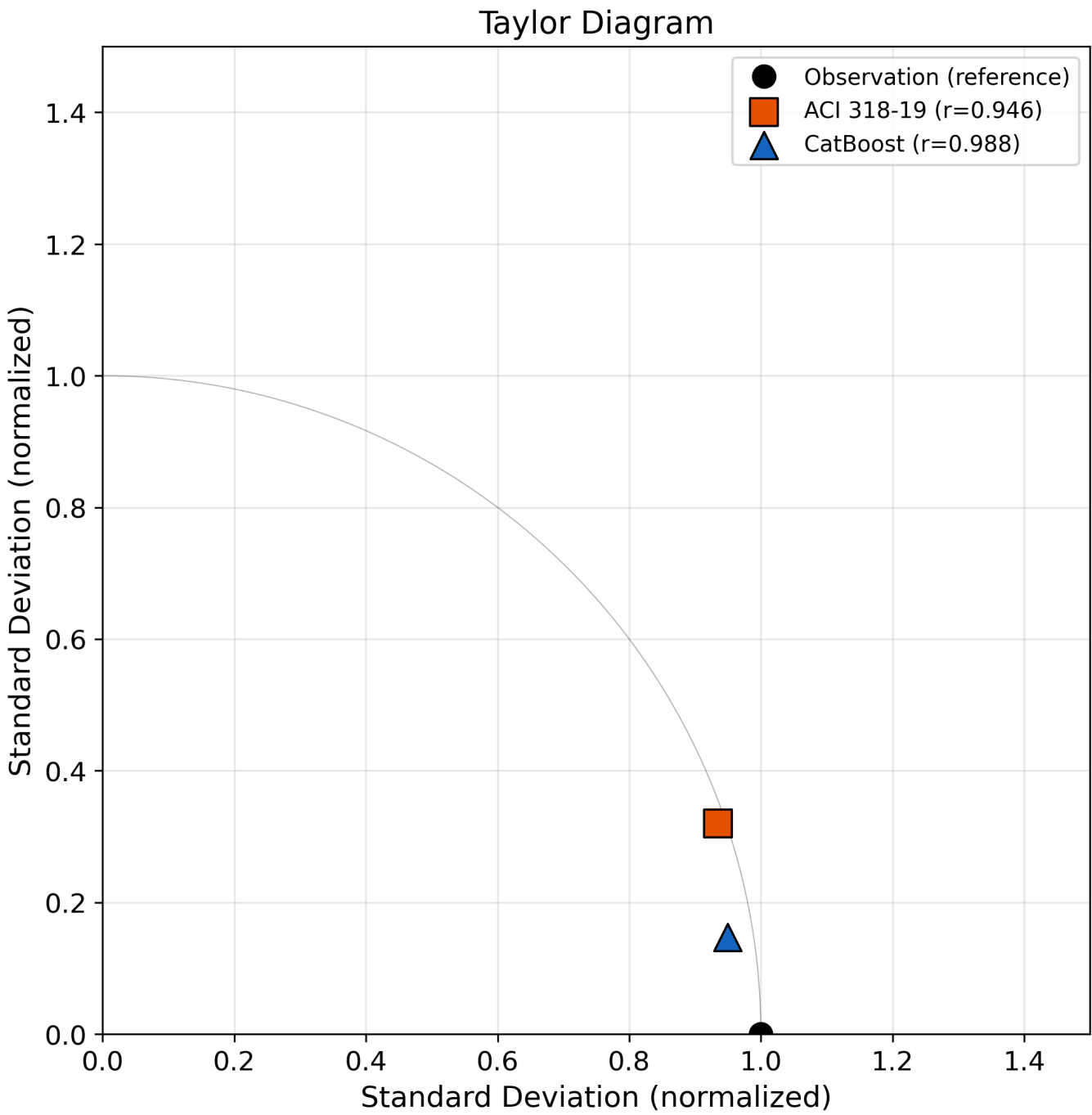


Fig8 Pysr Equation Scatter

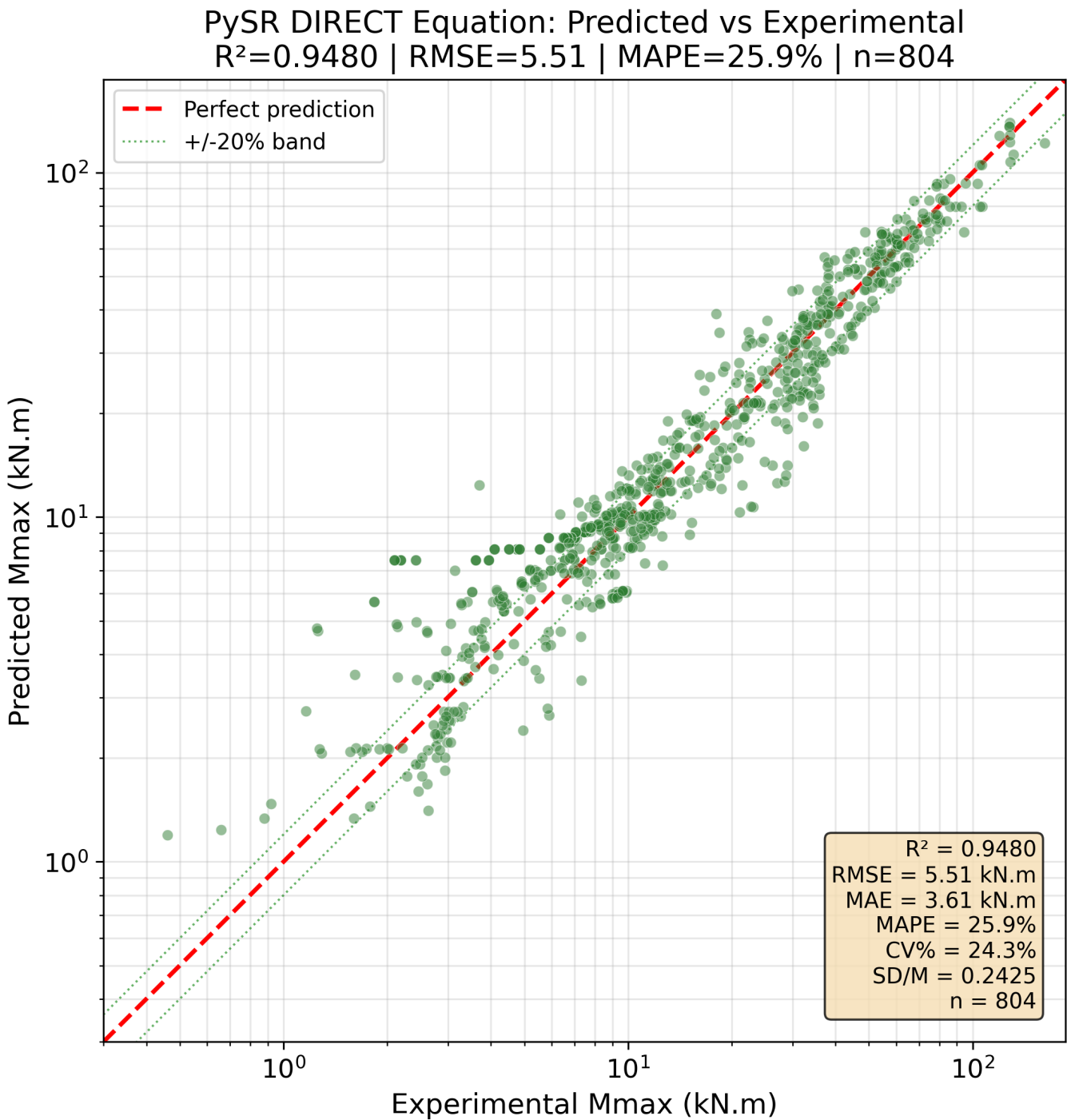
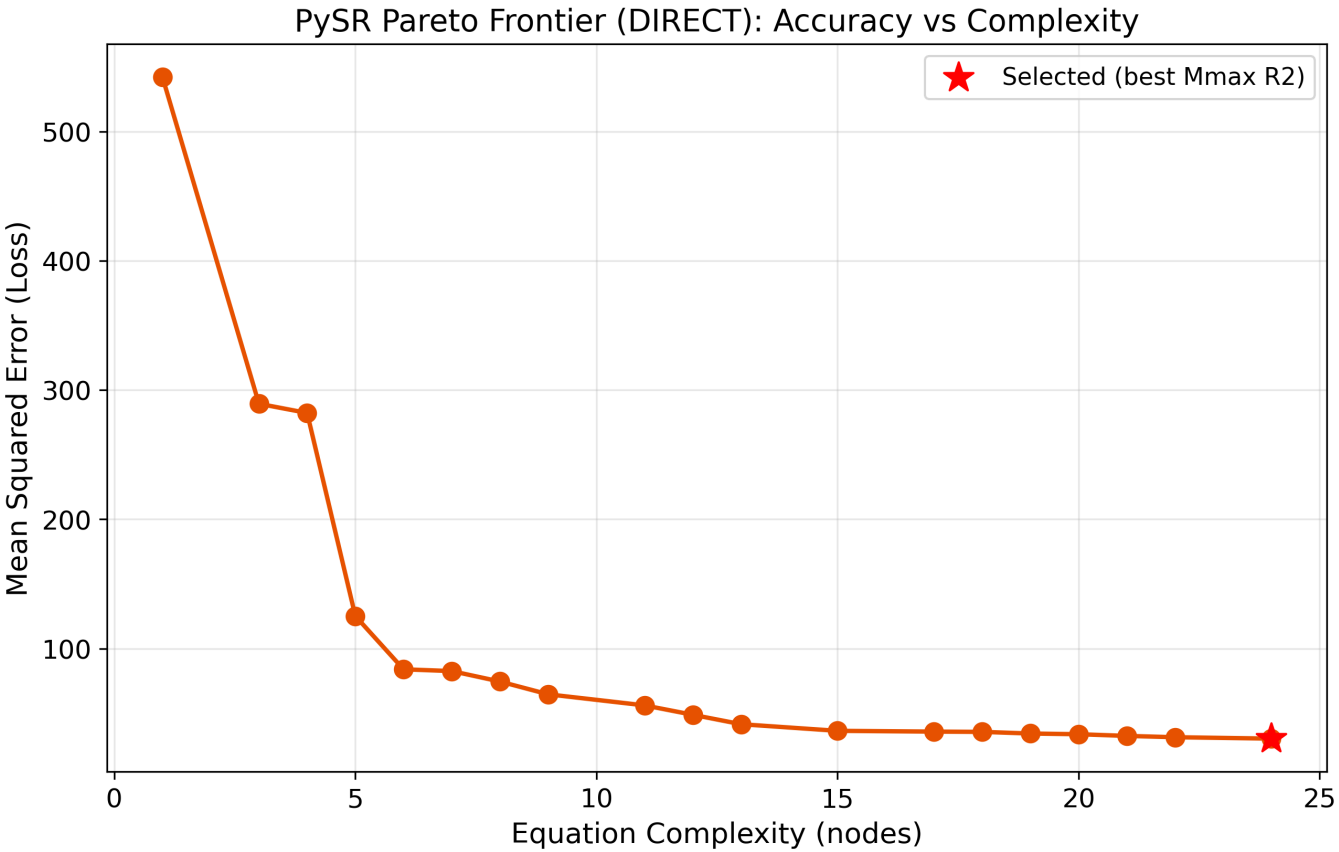


Fig9 Pareto Curve



Shap Beeswarm

SHAP Beeswarm Plot — Feature Impact on R(%) Prediction

